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Governing Enterprise AI

A Board and Senior Executive Reading of the OIA Reference Model.

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FOREWORD

Why this document exists

This document is written for the audience I sit on boards with: chief executives, board members, audit and risk committee chairs, and the lead independent directors who increasingly find AI on their committee agendas without a clear architectural vocabulary in which to discuss it.

The frameworks the board already uses — NIST AI RMF, ISO/IEC 42001, the EU AI Act, SEC and equivalent disclosure regimes, and the fiduciary doctrines that have governed enterprise risk for decades — all ask the board to govern AI. None of them tells the board what AI actually is, structurally, in language the board can use.

The Open Intelligence Architecture (OIA) Reference Model provides that structural vocabulary. This document reads the OIA for boards and the most senior executives — the people accountable for AI strategy and risk who do not need to architect the systems themselves, but do need to govern them.

It is brief on purpose. The full OIA body of work runs to forty pages of architectural specification plus audience-specific overlays for security, technology, and governance leadership. This reading distills what the senior-most accountable audiences should know — and the questions they should be asking — into something that can be completed before a committee meeting.

The architecture of AI accountability

What makes AI different, structurally

Three properties distinguish intelligent systems from the application architectures of the past. Each is the reason something in the architecture exists.

Persistence. Intelligent systems carry context, memory, and judgment forward across interactions. A conversation continues across sessions. An agent's knowledge of the organisation accumulates. A model's behaviour reflects its training history. The persistence is not optional — it is what makes the system useful. It is also what makes it harder to govern.

Autonomy. Intelligent systems act with degrees of authority humans did not pre-approve case-by-case. They take multi-step actions, invoke tools, and operate inside scopes of authority defined at deployment rather than at each transaction. They can act faster than humans can supervise them in the moment.

Consequence. Outputs and actions whose correctness must be demonstrable — and demonstrable against current threat reality, not the reality at the moment of deployment. A patched library, a swapped model, a fresh jailbreak: each changes correctness mid-flight. Architectural posture is time-bound.

These three properties are why AI is a different governance problem. Conventional application risk is largely point-in-time. AI risk is continuous.

The architecture in ten layers and seven cross-cutting concerns

The OIA describes enterprise intelligent systems as ten layers, numbered 0 through 9, with seven cross-layer concerns threading through them. The layers describe where the work happens. The cross-cutting concerns describe how the consequences propagate.

Ten layers, bottom up:

- **Layer 0 • Physical Compute.** Semiconductors, energy, facilities, geographic siting. *If the substrate is in the wrong jurisdiction, no amount of architectural sovereignty above can recover the posture.*

- **Layer 1 • Silicon Abstraction.** Hardware diversity behind portable runtimes and trusted execution primitives. *Verifiable trust starts here.*
- **Layer 2 • Sovereign Infrastructure.** Compute, storage, networking with sovereignty made explicit. *Sovereignty can't be retrofitted higher in the stack.*
- **Layer 3 • Agent Data Substrate.** The persistent, auditable, access-controlled data environment. *This is where AI's data exposure is either governed or isn't.*
- **Layer 4 • Model Training and Adaptation.** Pre-training, fine-tuning, and continuous adaptation. *Model weights are crown-jewel intellectual property; training data is where future regulatory liability is created or contained.*
- **Layer 5 • Inference and Retrieval.** Where models run and their trustworthiness becomes a live operational concern. *The model that decided is no longer necessarily the model deployed.*
- **Layer 6 • Context and Knowledge.** Retrieval, grounding, and continuous perception. *Adversaries who control the context window control the model output.*
- **Layer 7 • Orchestration and Workflow.** Multi-step agent action and tool invocation. *Three of the four agentic-era threat surfaces land here.*
- **Layer 8 • Continuity Fabric.** Cognitive state preserved through change. Recovery, rollback, post-incident integrity. *The architectural centrepiece — and where existing technology stacks are most incomplete.*
- **Layer 9 • Human and Browser Interface.** Where humans and intelligent systems interact. *If you can't tell humans from agents at every entry point, every other layer of governance can be bypassed at the door.*

Seven cross-cutting concerns, each anchored at one or more layers and propagating through several:

- **Security.** The widest concern; threads through every layer.
- **Sovereignty.** Three dimensions: jurisdictional, operational, energy. Established at Layer 2 and propagating upward.
- **Auditability.** The capacity to reconstruct what happened, after the fact. Cannot be retrofitted.
- **Identity.** Three actors require continuous trust governance: humans, agents, and foundation models themselves.
- **Energy and Environmental Concerns.** Architectural rather than just operational.

- **Provenance.** The lineage of information, models, decisions, and actions.
- **Verifiability.** Operations cryptographically proven, not contractually asserted.

Four named threats of the agentic era

The OIA names four threat surfaces specific to the agentic era. They share a pattern boards should notice: they do not live at the perimeter where prior security architectures expected attacks to land. They live inside the operational stack.

- **Mimicry** — telling humans from agents, and AI-generated content from human-authored, at every entry point.
- **Agent-in-the-middle** — adversarial intermediaries between agents and tools. The agentic equivalent of an SSL strip.
- **Agent spoofing** — agent identities inheriting trust they should not have, across contexts and lifetimes.
- **Model compromise** — the trusted model that is no longer trustworthy. Weight swap, alignment drift, jailbreak emergence.

These four are not a complete catalog of AI risk. They are the four that the agentic era has made canonical — the four that did not exist as named concerns when most boards last updated their AI risk treatment.

How the OIA composes with existing frameworks

The OIA is additive, not competitive. It addresses a domain that existing reference architectures and management frameworks were not designed to organise, while drawing on their intellectual conventions and complementing their coverage. The OIA provides the structural map within which NIST CSF, NIST AI RMF, MITRE ATLAS, MITRE D3FEND, OWASP Top 10 for LLM Applications, the Confidential Computing Consortium body of work, and ISO/IEC 42001 compose. The EU AI Act's evidence requirements map directly to the OIA's cross-layer spans — auditability, provenance, identity, sovereignty, and verifiability are exactly the artefacts the Act requires evidence for. The OIA's contribution is shared structural vocabulary. The frameworks then say what good looks like inside that vocabulary.

PART TWO

Where boards should focus

Four architectural arguments distilled from the OIA. Each converts something the board has been hearing about as a technical concern into a structural question the board is uniquely positioned to engage with.

The state-holding question

Where does our enterprise's cognitive state actually live, and can we recover when something goes wrong?

The OIA holds enterprise cognitive state in two architecturally distinct layers, deliberately bracketed around the operational stack. Layer 3 (Agent Data Substrate) is the persistent foundation. Layer 8 (Continuity Fabric) is the active layer at which state is preserved through change.

The board question is not whether the enterprise has backups. The board question is whether, when an agent acts autonomously and the action turns out to be wrong, the enterprise has the architectural capacity to reconstruct what happened, to identify which model decided, and to recover the state that existed before the action. The OIA names Layer 8 as the architectural centrepiece — and as the layer where the existing technology stack is most consistently incomplete. The board's question to management: are we relying on agent framework vendors and model providers to hold our cognitive state, and what is our exposure when those vendors change?

The operational stack is the new attack surface

If three of four named threats live inside the operational stack, where is our security investment weighted?

Three of the four named threats of the agentic era originate or propagate through the operational stack — not at the perimeter. Mimicry lands at the human interface. Agent-in-the-middle and agent spoofing live in orchestration. Model compromise originates at inference.

The board implication: AI security investment cannot be weighted toward perimeter hardening. The threats are where the work happens. An enterprise that has hardened the network and identity perimeter but has not addressed semantic-

layer threats inside the operational stack has defended against an earlier era's threats.

Verifiable trust replaces contractual trust

Where are we relying on contractual trust where the workload now requires verifiable trust?

Trust in enterprise systems has historically been contractual — vendors promise, customers verify through audit, and the system works because the promises are large enough to enforce. AI breaks this model in two places. First, foundation model providers can change behaviour under the hood between procurement and incident. Second, in sovereign and regulated workloads, the operator must not need to be trusted at all.

The OIA names a new cross-cutting concern — Verifiability — that captures the property that operations can be cryptographically proven rather than contractually asserted. Anchored in hardware-attested execution at Layer 1, operationalised through Layers 2, 4, 5, and 8. For boards in regulated industries (finance, healthcare, defence) and for sovereign workloads, verifiable trust is becoming the requirement rather than the option.

Three actors need governance

Are we governing all three actors, or just the humans?

Identity has historically governed one class of actor: humans. AI introduces two more — agents (which act on behalf of humans or organisations) and foundation models themselves (which can no longer be treated as static products because their behaviour changes under the hood). The OIA's contribution is naming these three actors as one governance problem at three architectural layers — humans at Layer 9, agents at Layer 7, foundation models at Layer 5.

The board implication: the existing enterprise identity infrastructure was not built for non-human actors. Agents inherit trust they should not have when they are not scoped, attested, and behaviourally baselined. Foundation models change behaviour and cannot be governed by procurement-time evaluation alone.

PART THREE

The questions to ask

The board's leverage is in the questions it asks the executives accountable for AI. These are not technical questions in the sense of requiring specialist knowledge to ask; they are architecture-level questions that surface posture rather than implementation.

To the Chief Information Security Officer

- Can we tell humans from agents at every entry point — and is our deepfake defence keeping up?
- Are we continuously scoring the trustworthiness of every model we use, against the threat reality of this week and not last quarter?
- Are agent identities scoped, attestable, and prevented from inheriting trust they should not have?
- Can we trust the chain between every agent and every tool, including MCP servers, at machine speed?
- What is our recovery plan if an autonomous agent has mis-acted and the audit chain is broken because the model that decided is no longer the model deployed?
- How are we protecting model weights against the full threat landscape, including nation-state actors where applicable?

To the Chief Information Officer or Chief Technology Officer

- Where does our stack end and our partners' stack begin?
- What is our exposure to GPU concentration — and is that exposure architectural or just contractual?
- How are we governing the agent frameworks we have in production — and the new ones coming?
- Where do we have de facto multi-cloud AI deployment without de jure multi-cloud architecture?
- What does our continuity fabric look like — the layer above the agent frameworks that owns our cognitive state when they change?

To the General Counsel and Chief Compliance Officer

- Where does the EU AI Act apply to systems we are deploying or relying on, and have we classified them correctly?
- What is our liability exposure when an agent acts on our behalf?
- What is our IP exposure in training data, fine-tuning data, and model weights?
- Are our contracts with AI vendors built for the agentic era — including liability for model behaviour changes, weight protection guarantees, and continuity obligations?
- Are we ready for the first serious regulatory inquiry about material AI risk disclosure?

To the Chief Financial Officer

- What is our total AI capital deployment, and what is our exposure to single-vendor or single-region concentration?
- How are we capitalising AI infrastructure investments — and how should we be?
- What is our material AI risk for disclosure purposes?
- How is FinOps thinking about AI when token costs, GPU-hours, and training amortisation break the categories our finance organisation thought were settled?

To the Chief Executive Officer

- Are we governing AI strategy, or receiving reports about it?
- Where is AI changing our business model fundamentally — and where is it enhancing existing operations?
- What is our position on sovereign AI and on attested computing — defaults, regulated-workload exceptions, or unsettled?
- Who, in our organisation, owns the architectural conversation about AI — and is that person at the right level?
- What signal would a serious AI failure send to our customers, our regulators, and our employees — and are we prepared for that signal?

PART FOUR

What good governance looks like

Five postures distinguish boards that are governing AI from boards that are receiving reports about it.

Govern strategy, do not receive reports

The board's contribution is direction, not approval. Receiving regular AI updates from management is necessary but not sufficient. The board's question is whether AI investment, exposure, and posture serve the strategic direction the board sets — and whether the strategic direction itself has been adjusted for what AI now makes possible, and what it now requires.

Ask architecture-level questions, not implementation questions

"Where in the architecture does this concern live?" is a stronger question than "what product are we using?" Architecture questions surface posture; implementation questions surface vendor choices. Boards have leverage at the posture level.

Refuse vendor framing

No vendor's product map is the architecture. The OIA and similar reference models exist precisely so that boards and customers do not have to organise their thinking around vendor taxonomies. When a vendor describes "the AI platform," the board's question is which OIA layers it covers, and which it leaves to the enterprise or to other partners.

Compose with existing governance, do not replace it

AI is not a separate governance regime. It is a new structural domain that the board's existing governance frameworks must extend into. NIST AI RMF, ISO/IEC 42001, the EU AI Act, and the existing fiduciary frameworks all extend; none needs to be replaced. The OIA gives the board the structural vocabulary in which those frameworks now apply.

Accept that some questions are still genuinely open

The board does not need management to have answered every question. The board needs management to have named every question, to know which are answered and which are open, and to be making progress on the open ones. The OIA helps by naming the open architectural questions explicitly — at Layer 8 (continuity primitives that work across agent frameworks), at Layer 7 (agent identity governance at architectural scope), at Layer 5 (continuous model trust scoring), and across the Verifiability span (verifiable rather than asserted trust as an enterprise default rather than a regulated-workload specialty).

CLOSING

The posture question

The board's contribution to AI is governance. Governance requires architectural literacy. Architectural literacy requires shared vocabulary that does not depend on any vendor's product map or any single role's working language.

The Open Intelligence Architecture is offered to the industry as that shared vocabulary, published as an open reference under consideration for adoption by the Agentics Foundation. The architecture itself, the lens system that translates it for executive audiences, and the broader body of work are open for engagement, critique, and refinement.

This document is the board's reading of that work — short enough to be read before a committee meeting, substantive enough to inform what the board asks management between meetings, and grounded in the architectural facts that the existing governance frameworks assume but do not supply.

The next conversation is the one the board has with the chief executive about what posture the enterprise is taking on each of the architectural questions named here. That conversation does not require board members to be AI architects. It requires the board to be willing to govern at architectural depth — and to use the shared vocabulary to do so.

— *Mark Templeton*